

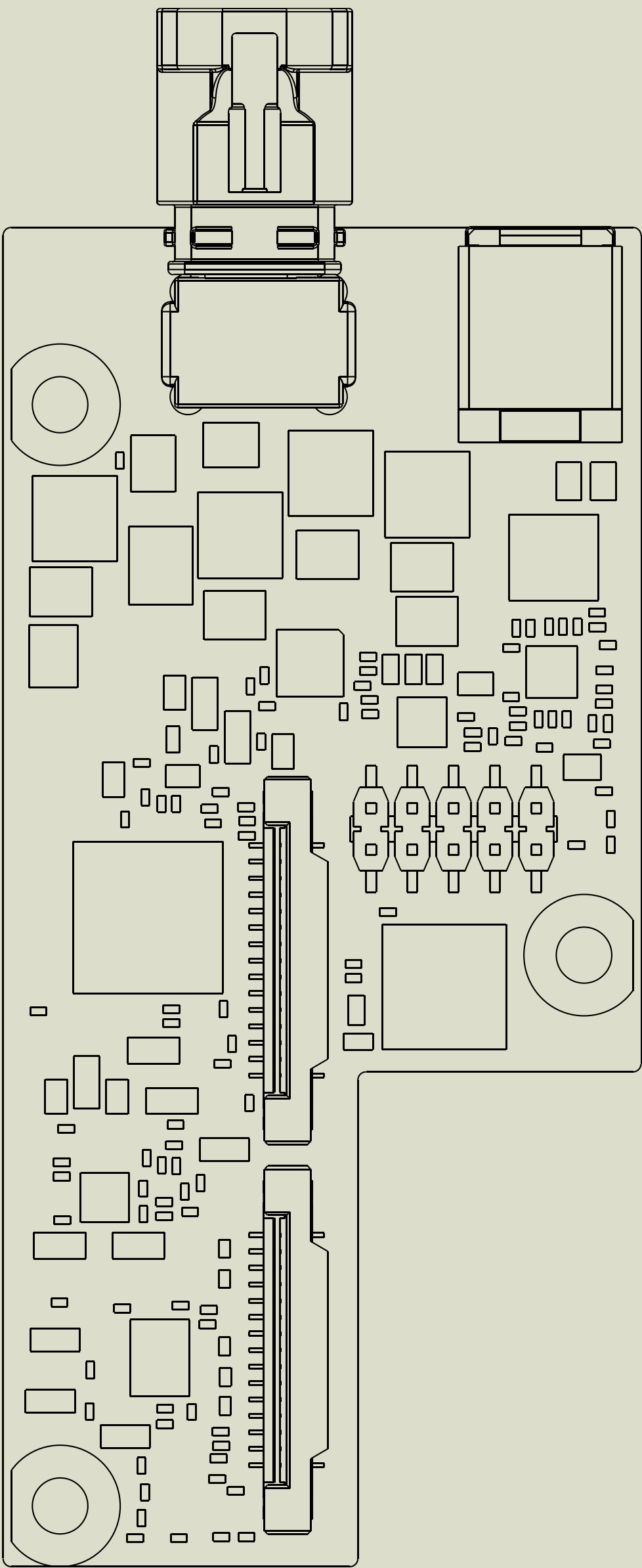
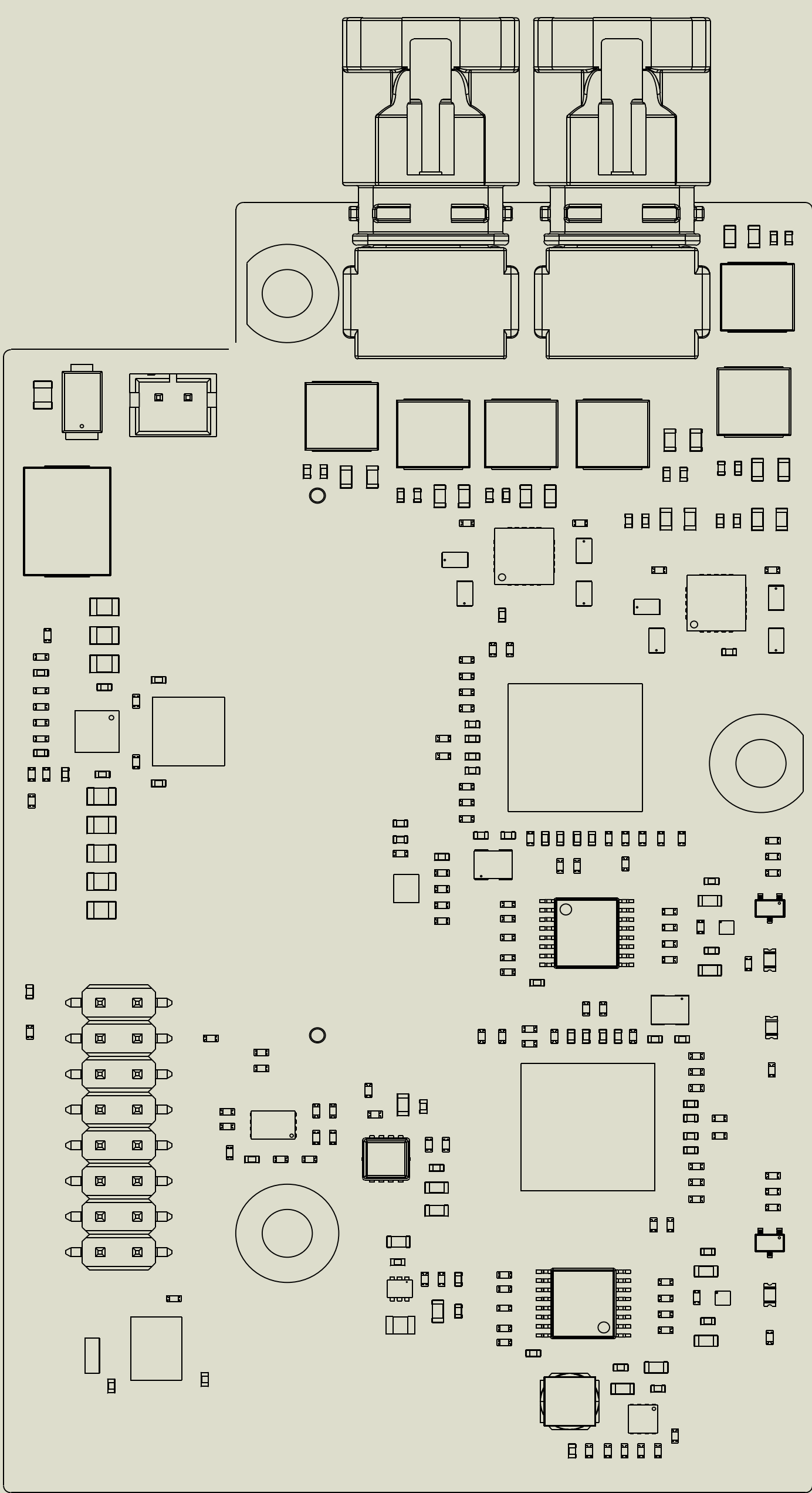
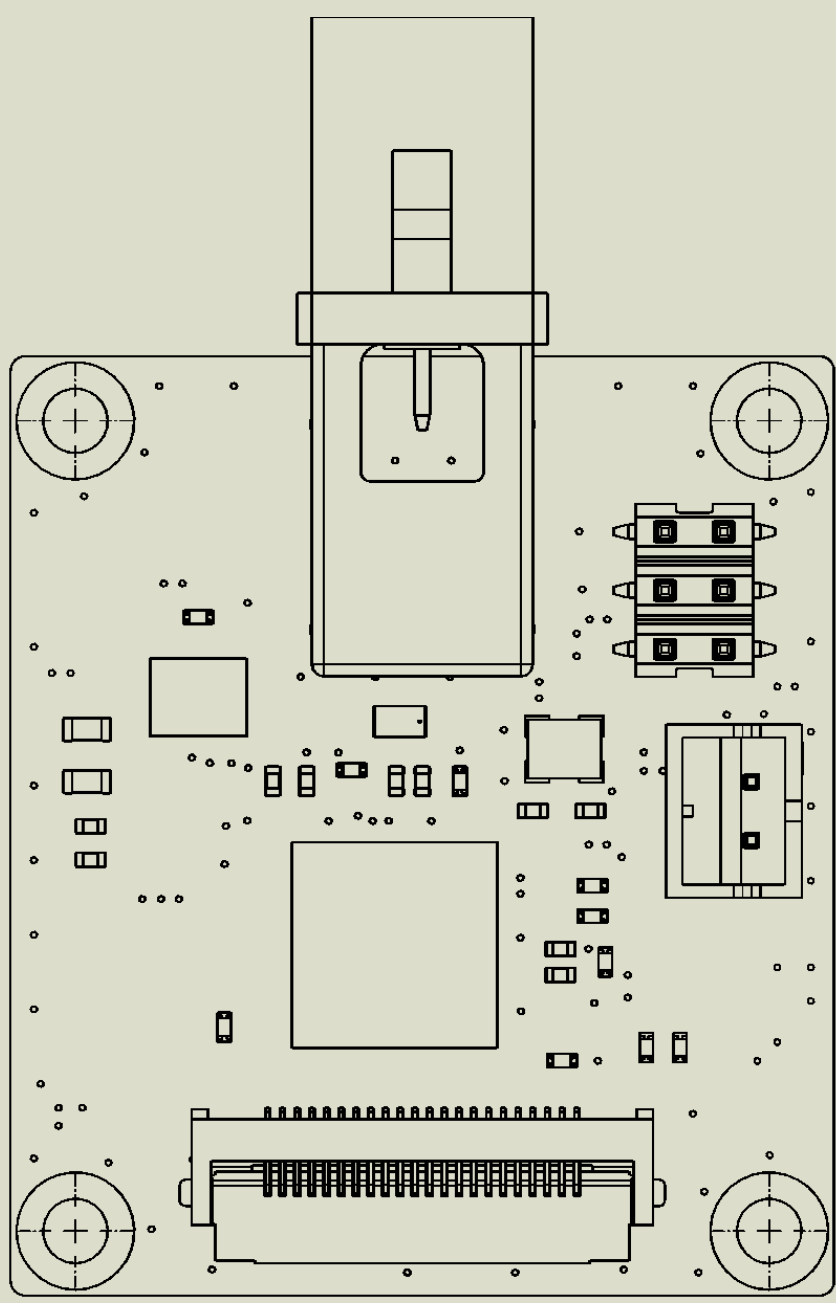
Reference Design Guide

GMSL2 Capture Card

For compatibility with Stereolabs ZED X cameras

Based on ZED X Drivers V1.3

April 2025 - Revision 1.3



General Informations

To unlock the full potential of the firmware and connect a maximum number of ZED X cameras to the host system, the following design guidelines should be followed. A different architecture would result in reduced performance in connectivity and video bandwidth.

Due to multiple components included in the ZED cameras (IMU, EEPROM and optic sensors), two main parameters are to be taken into consideration while designing your GMSL capture cards : the i2c root and the MIPI CSI lanes root.

I2C

ZED X cameras are composed of various components that use I2C to communicate with the ECU. To ensure full compatibility, it is recommended to use one I2C channel per deserializer. This can be achieved using one of the following approaches:

1. TCAs

A TCA can create multiple I2C channels muxed in one I2C port. This allows setting deserializers and cameras on different channels. This configuration allows multiple deserializers to be connected to the same I2C port. However, having numerous cameras on a single I2C port may result in bandwidth limitations, particularly when accessing IMU data.

2. Multiple I2C ports

Similarly, using multiple physical I2C ports enables separation of channels without relying on a TCA. With a single deserializer per I2C port, communication with the IMUs and other internal components of the ZED X camera is more reliable.

To ensure proper I2C communication with each camera, a maximum of four ZED X cameras should be linked per I2C port.

MIPI CSI lanes

The goal when rooting MIPI CSI lanes on the capture card is to allow sufficient bandwidth for camera streaming. A maximum of four video streams can be streamed through a single MIPI CSI port (whether configured with two lanes or four), meaning that not routing enough MIPI CSI lanes will result in fewer connectable cameras. For optimal integration, it is advised to have one port (2-lanes or 4-lane) per GMSL input.

Jetsons limitations

Depending on the Jetson used, limitations may arise due to the number of available MIPI CSI ports. The AGX series provides 16 MIPI CSI lanes, allowing support for up to 8 GMSL ports (using MAX9296 or MAX96712 alike) at maximum frame rate. The NX and Nano modules offer only 8 MIPI CSI lanes. With 8 GMSL ports (using MAX9296 or MAX96712 alike), bandwidth limitations will occur, resulting in reduced frame rates as previously explained.

MAX9296

The MAX9296 deserializer can run either as a 2-lanes or a 4-lanes configuration per port. A 2x2 lane configuration (on a single port) is not possible. Therefore, the available options are to either use a single MIPI CSI port (2 lanes), which may result in bandwidth limitations, or utilize both MIPI CSI ports.

On the mono capture card (Figure 1), only one MIPI CSI port was used, which was sufficient since only a single GMSL input was connected to the deserializer.

In contrast, the ZED Box Mini (Figure 2) features two GMSL ports linked to a single MAX9296 deserializer. To avoid bandwidth constraints in this configuration, both MIPI CSI ports were utilized.

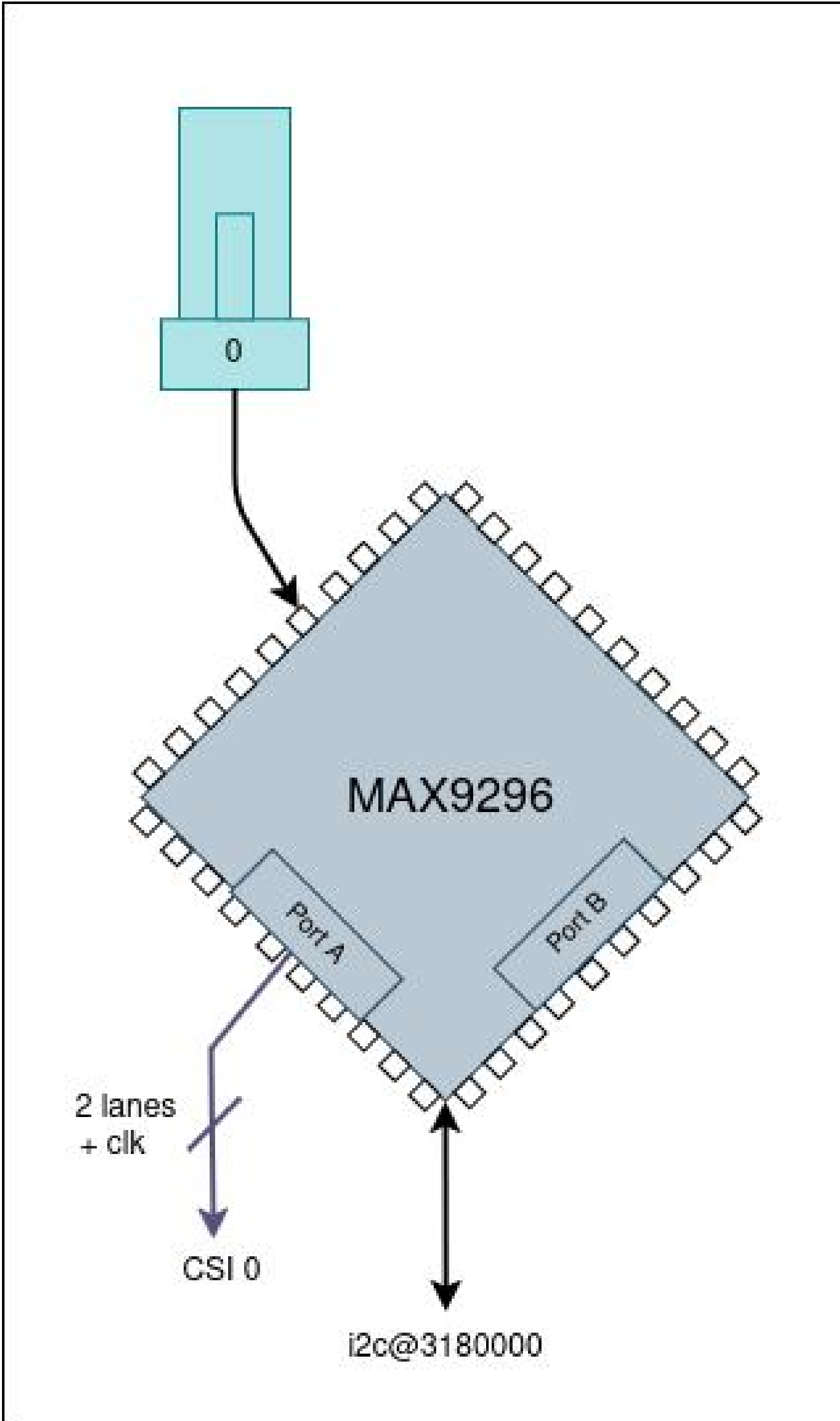


Figure 1: ZED Link Mono deserializer configuration

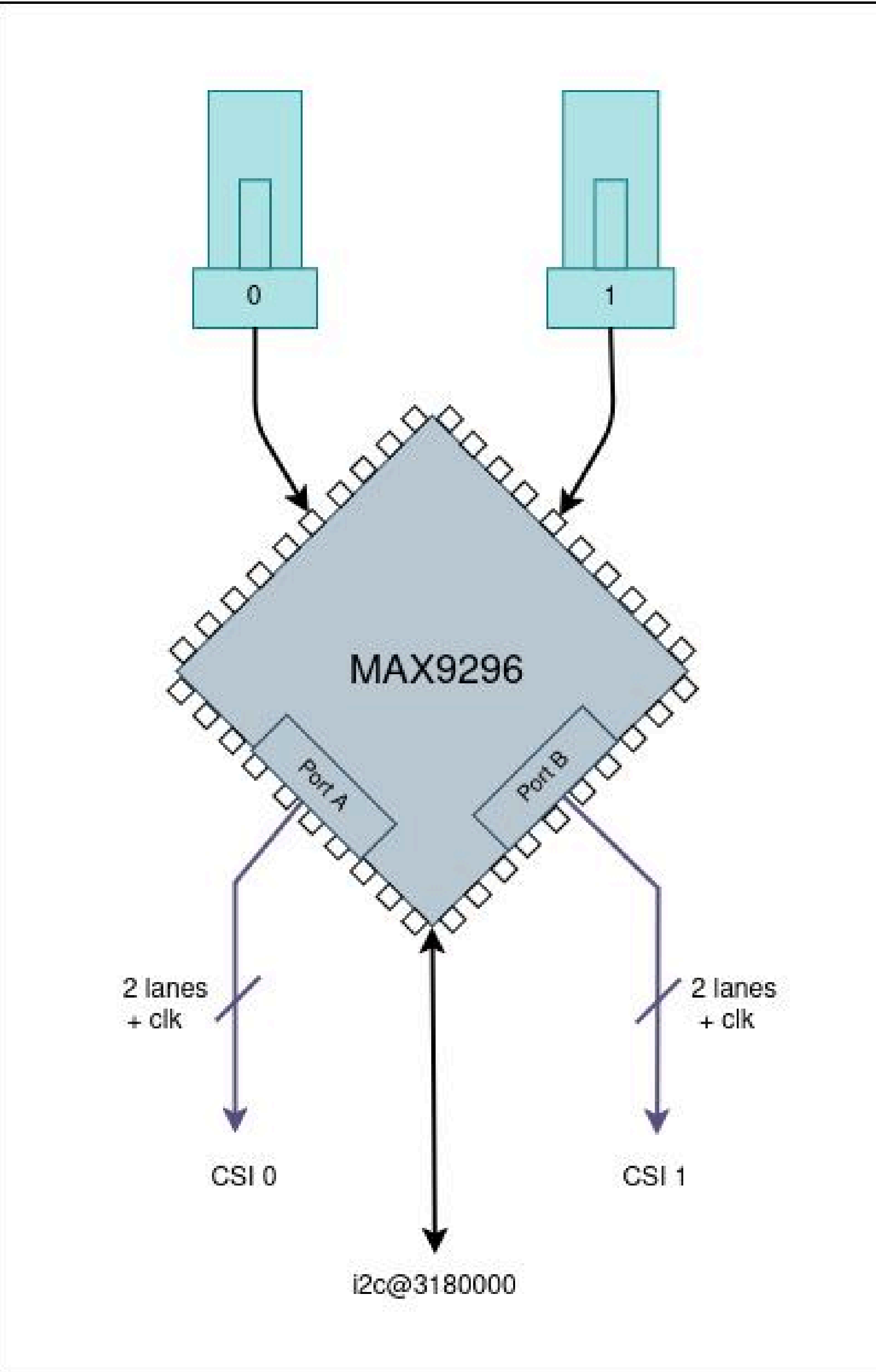


Figure 2: ZEDBox Mini deserializer configuration

In a multi-deserializer setup (Figure 3), each deserializer must be assigned to a separate I2C channel. This can be accomplished by utilizing TCA multiplexers, different I2C ports, or a combination of both, as previously explained.

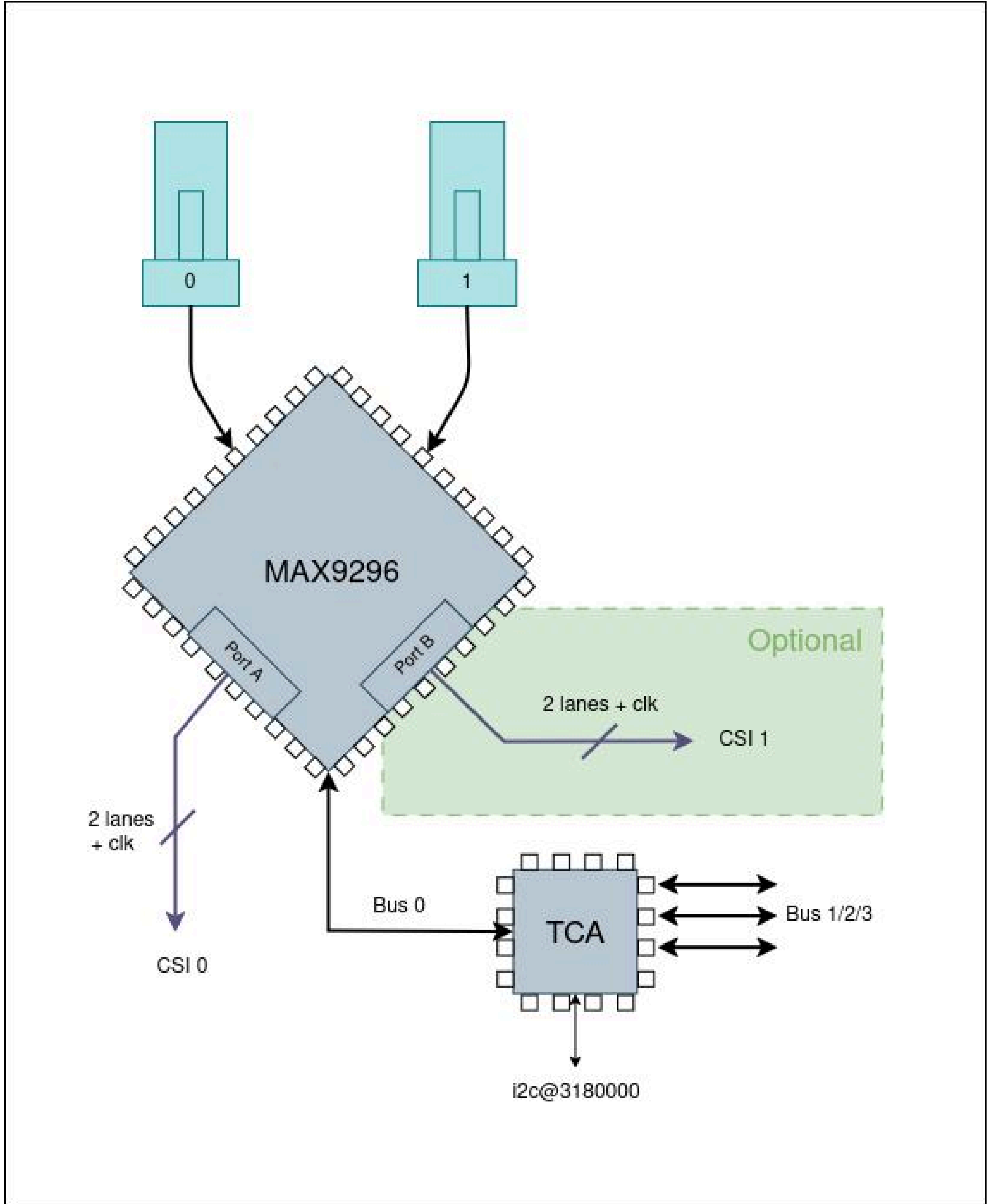


Figure 3: Multiple MAX9296 reference design

MAX9296 bandwidth capabilities	
Ports	2 ZED X
A	HD1200@30FPS
A & B	HD1200@60FPS

Table 1: MAX9296 bandwidth capabilities

MAX9296 Architecture examples

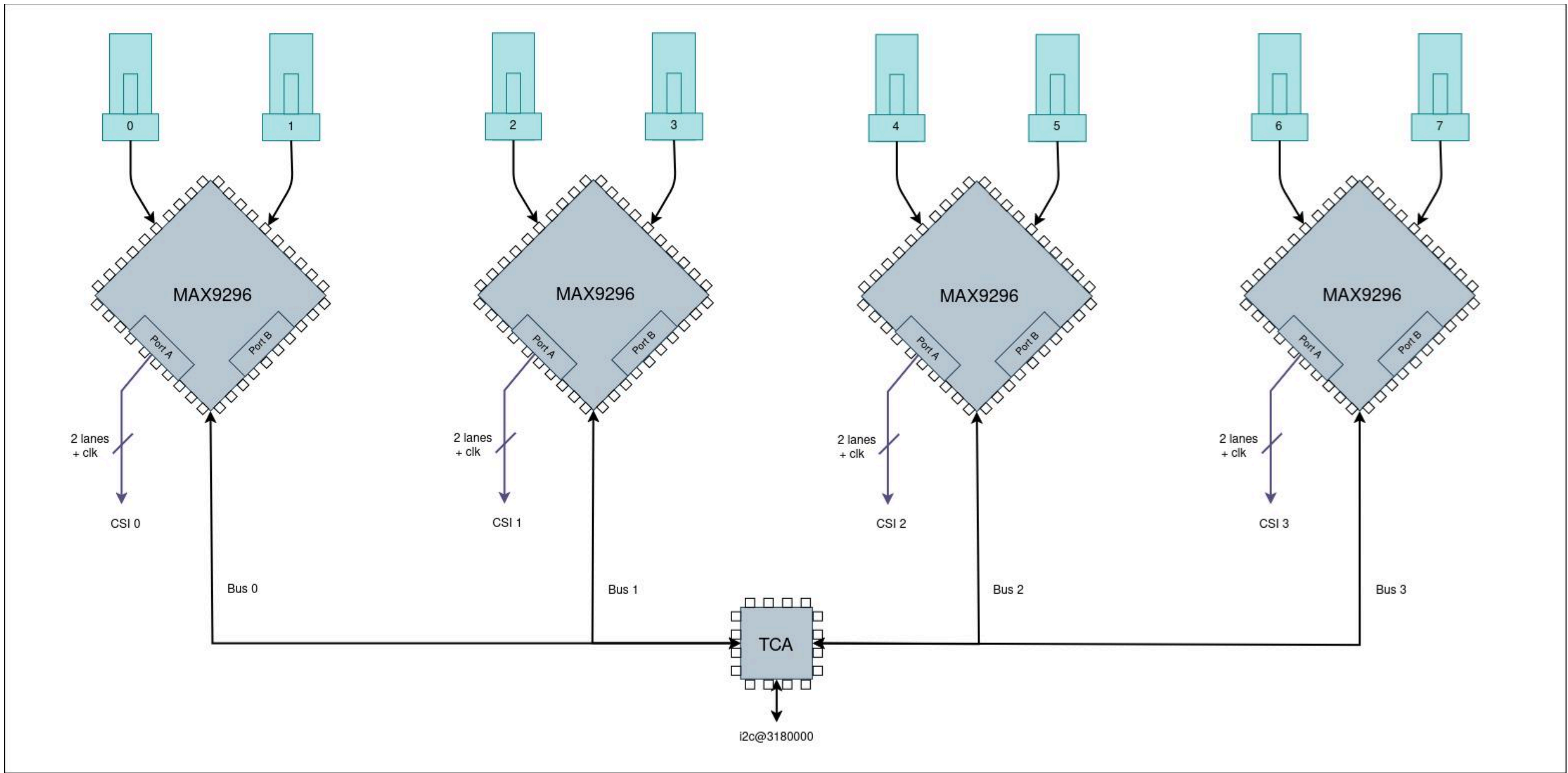


Figure 4: MAX9296 architecture A
Uses one I2C port muxed in 4 distinct I2C channels.
Each deserializer only has one out of two MIPI ports routed

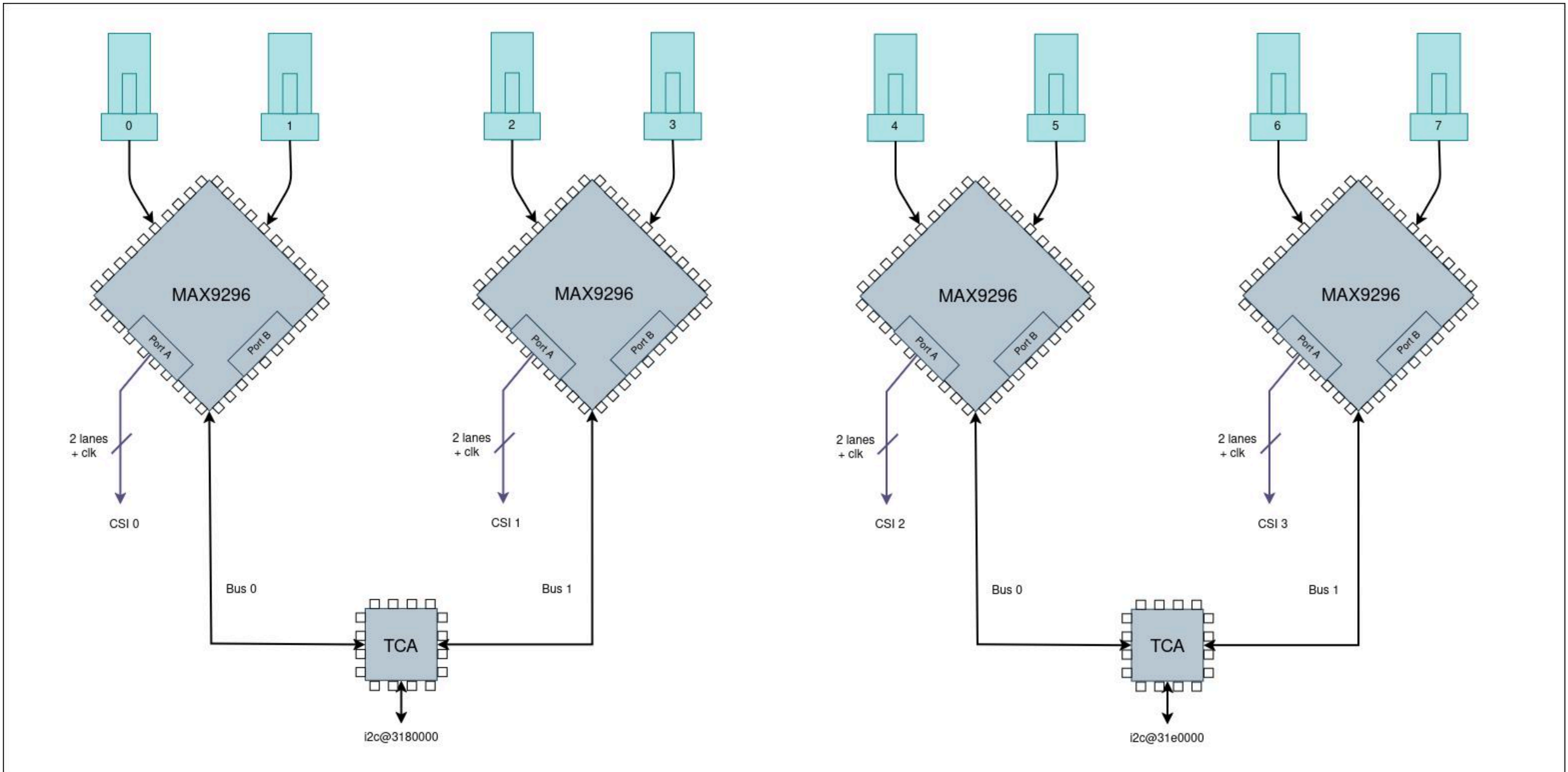


Figure 5: MAX9296 architecture B
Similar to architecture A, but with two I2C ports for more I2C bandwidth.

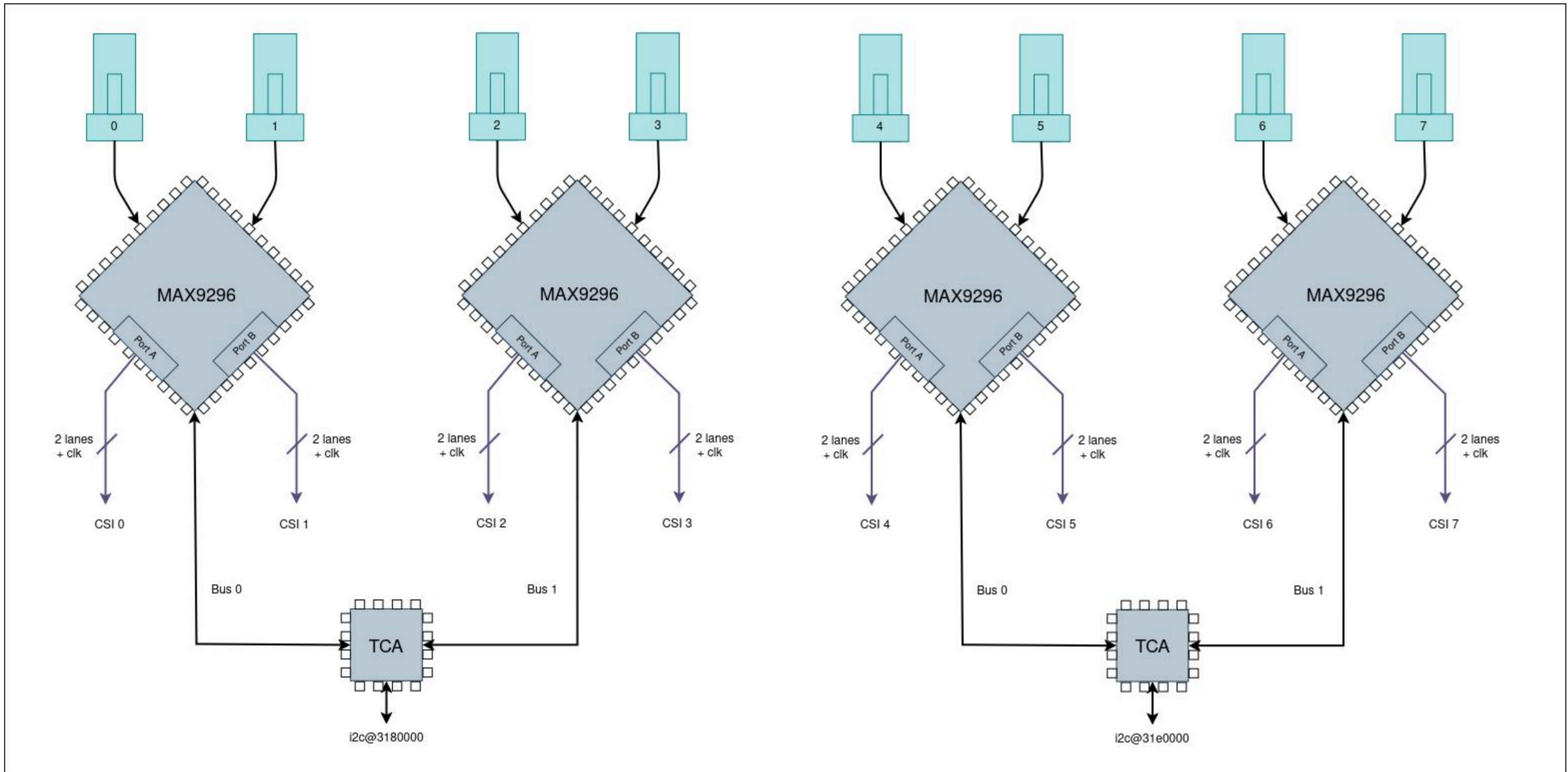


Figure 6: MAX9296 architecture C
Similar to architecture B, but with all MIPI ports for more MIPI bandwidth.

MAX9296 architecture capabilities comparaison			
Architecture	I2c Communication	Bandwidth	Compatibility
A	⚠️ Poor I2C communication	8 ZED X at HD 1200@30fps	AGX, NX, Nano
B	✅	8 ZED X at HD 1200@30fps	AGX, NX, Nano
C	✅	8 ZED X at HD 1200@60fps	AGX

Table 2: MAX9296 architecture capabilities comparison

MAX96712

Disclaimer: the current firmware (v1.3.0) supports only 2-lane configurations on the MAX96712. 4-lane configurations are not supported at this time. To ensure the correct MIPI configuration, all clocks must be routed properly.

The MAX96712 is a 4-GMSL entry deserializer, commonly used in Stereolabs dual and quad capture cards. It features four pairs of MIPI lanes, which can be configured as 2×4 lanes or 4×2 lanes. In the current firmware, the 4×2 lane configuration is utilized.

Each deserializer on the capture card is assigned to its own I2C port, facilitating reliable I2C communication with all cameras. While TCAs are not required, they provide the benefit of isolating cameras on different I2C channels, which aids in identification and debugging during multi-camera configurations. With proper routing of all MIPI lanes, the system can fully leverage the hardware, enabling HD1200 streaming at 60 fps from up to 8 ZED cameras simultaneously.

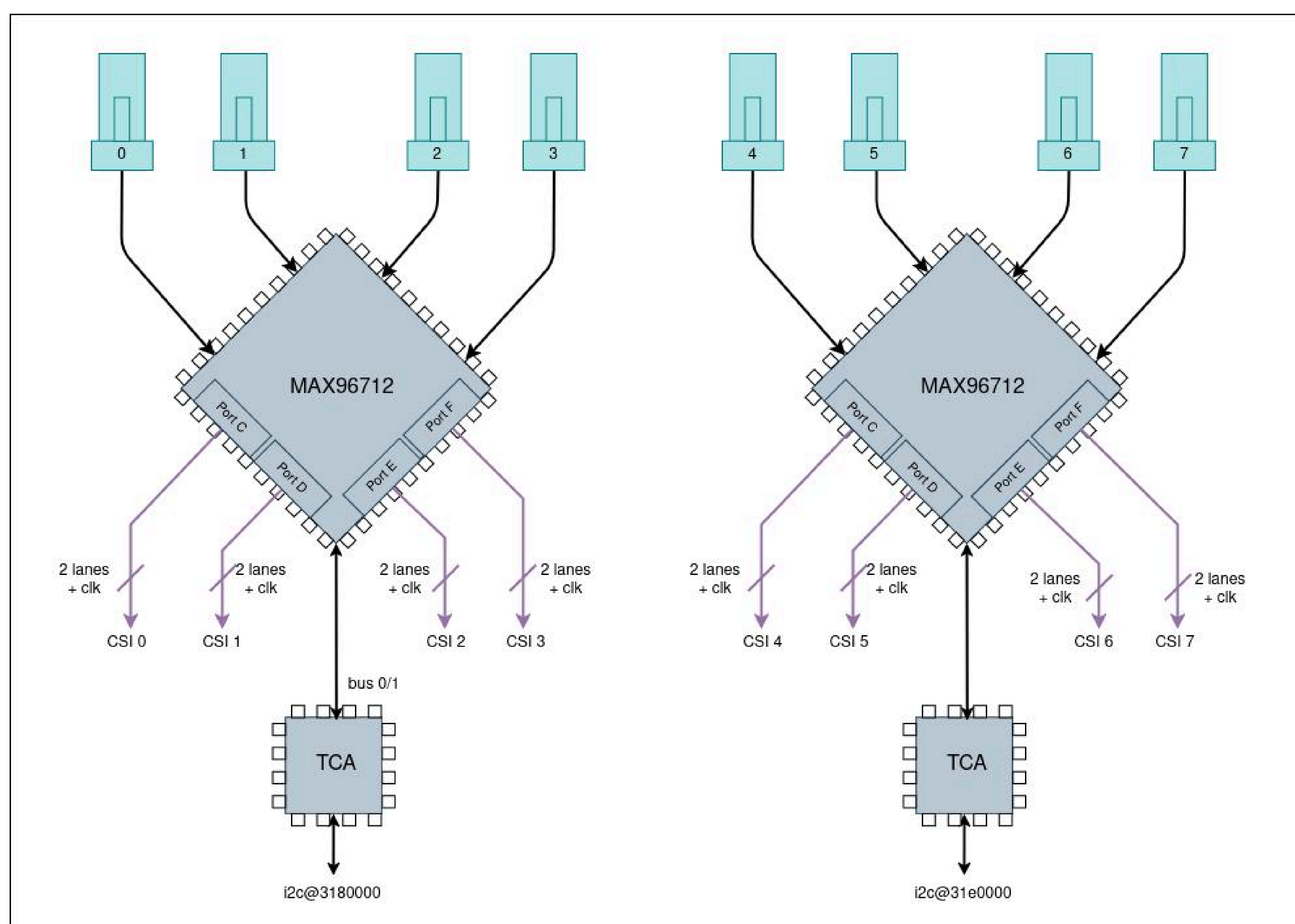


Figure 7: ZEDLink Quad architecture

As a regular design reference for MAX96712, it is necessary to have one I2C channel or port per deserializer for identification. Due to the high volume of I2C communications, using one i2c port per deserializer (see Figure 7) is strongly recommended.

MAX96712 bandwidth capabilities	
Lanes	2 ZED X
2 x 2	HD1200@30FPS
4 x 2	HD1200@60FPS

Table 3: MAX96712 bandwidth capabilities with two ZED X

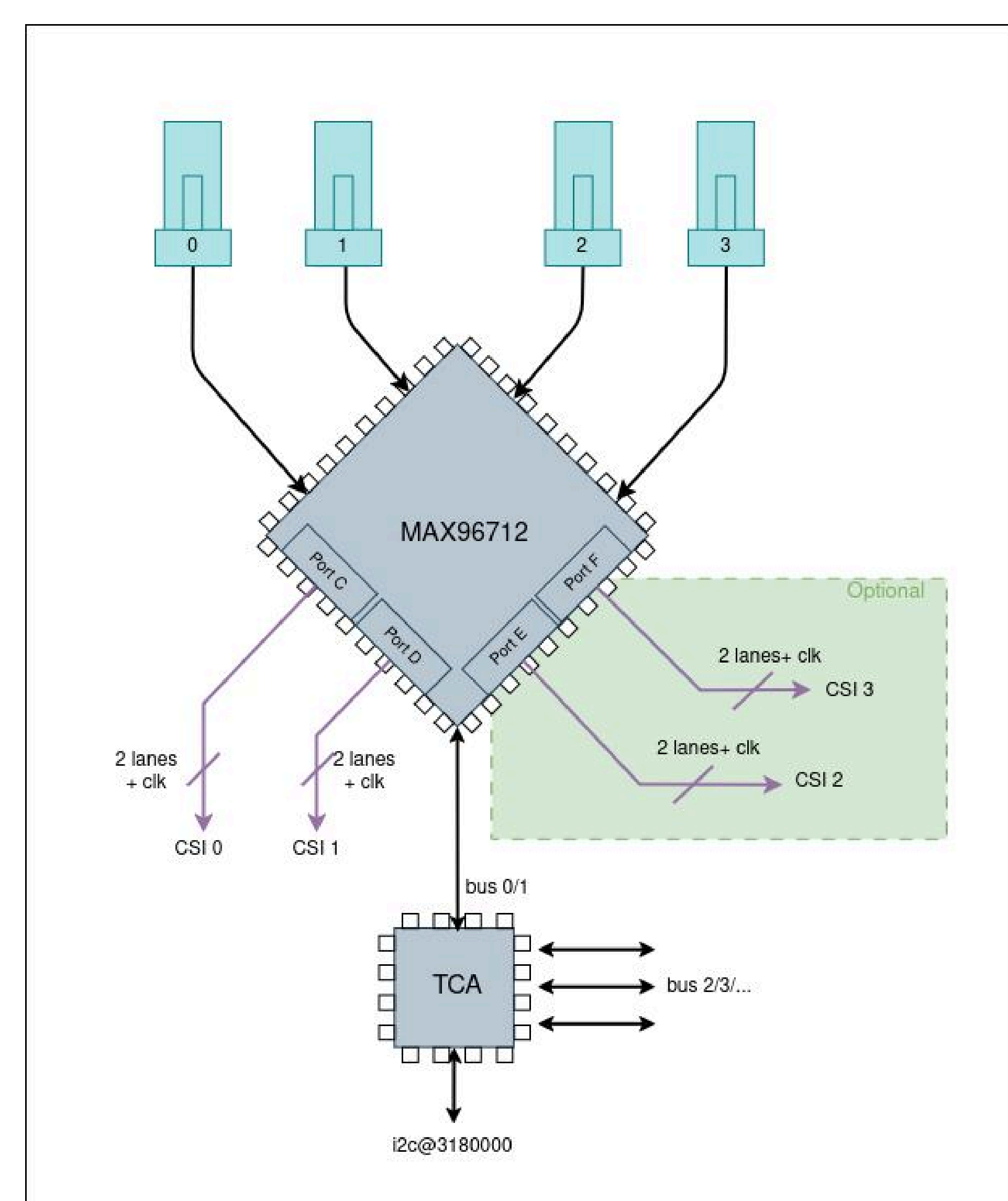


Figure 8: MAX96712 reference design

MAX96712 Architecture examples

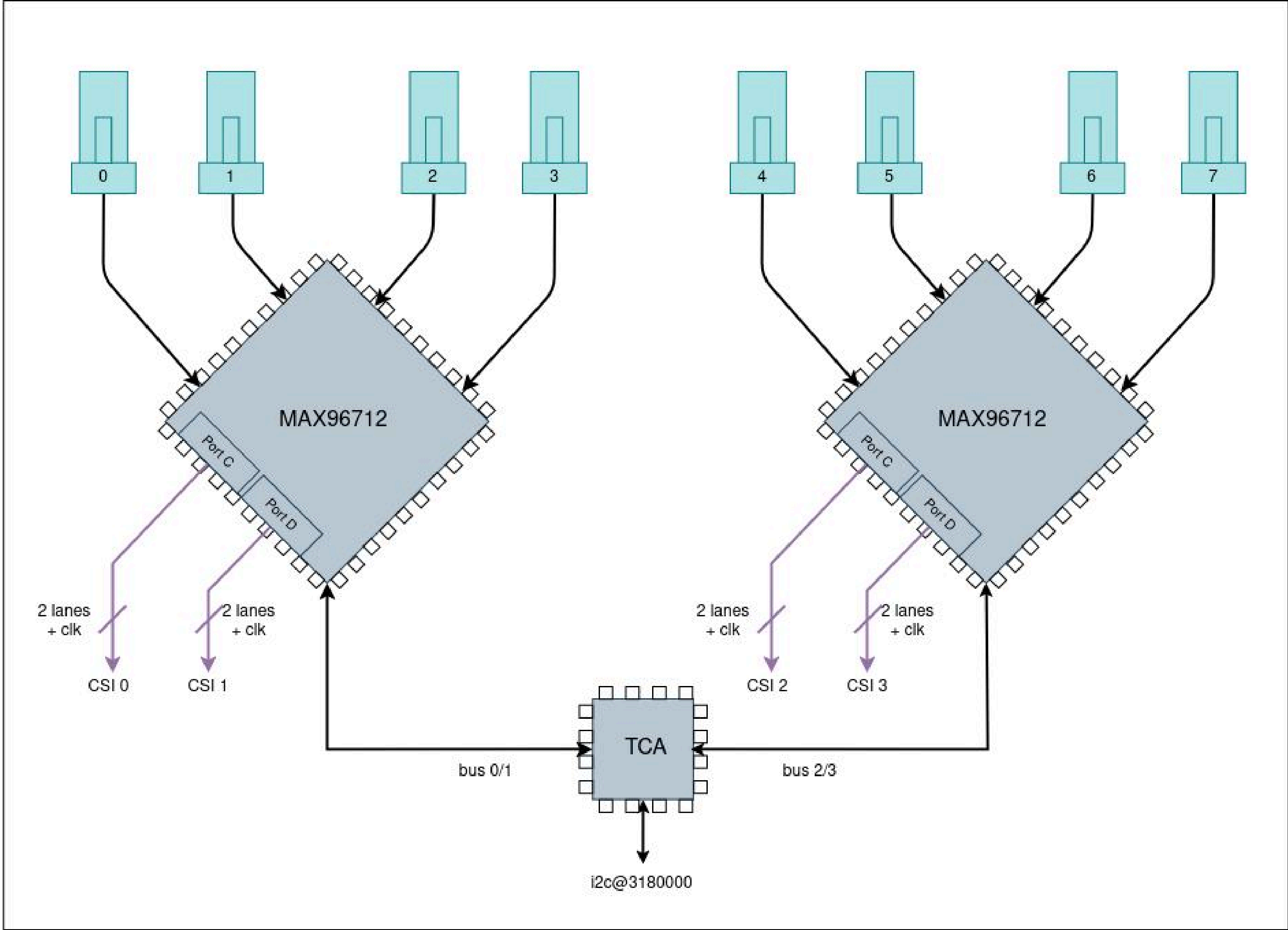


Figure 9: MAX96712 architecture A
Uses one I2C port demuxed thanks to a TCA in two I2C channels (up to two per deserializer). Only 2 out of 4 MIPI ports used.

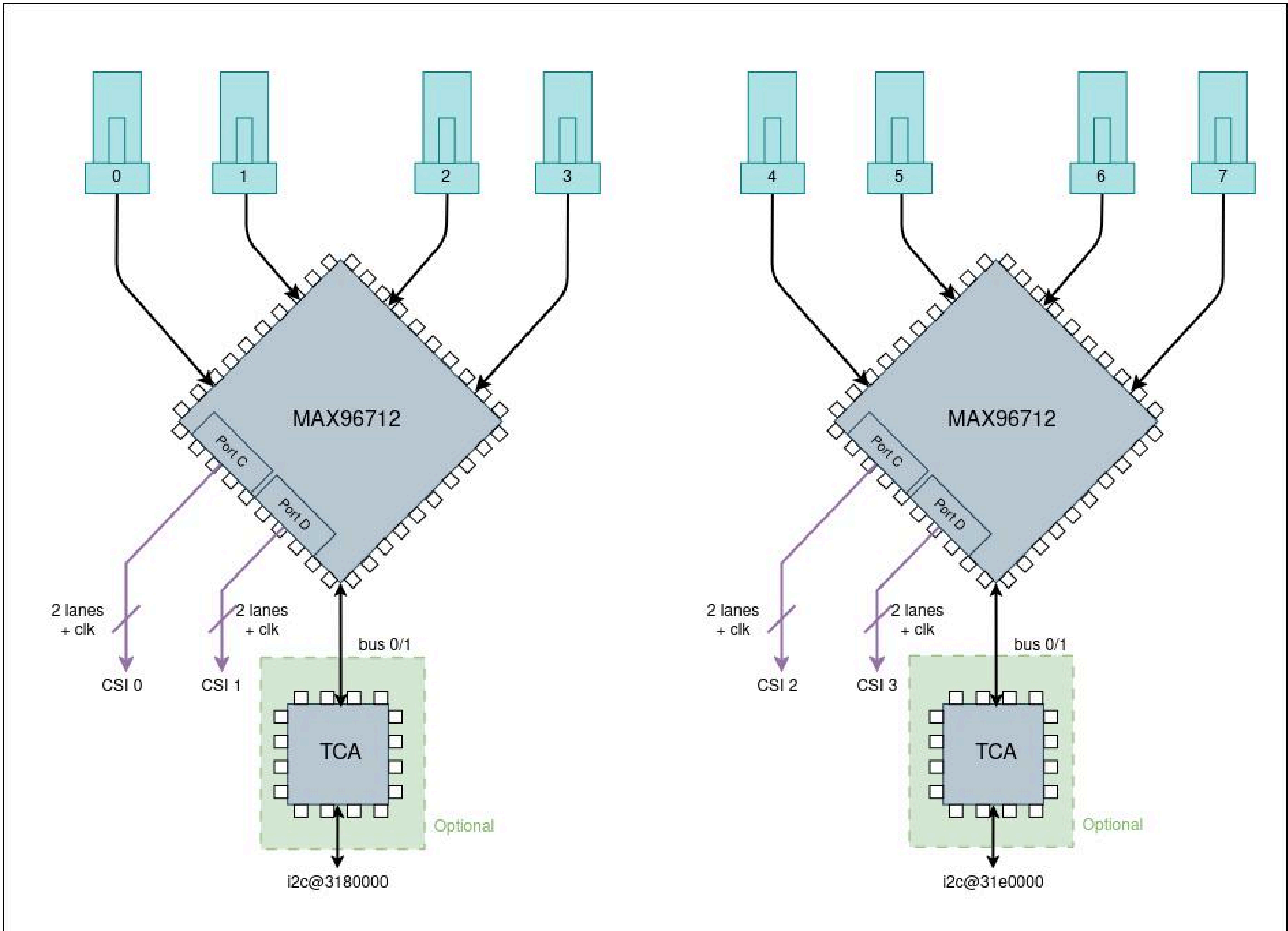


Figure 10: MAX96712 architecture B
Each deserializer has its own I2C port. TCA can help manage i2c address attribution. Only 2 out of 4 MIPI ports are used.

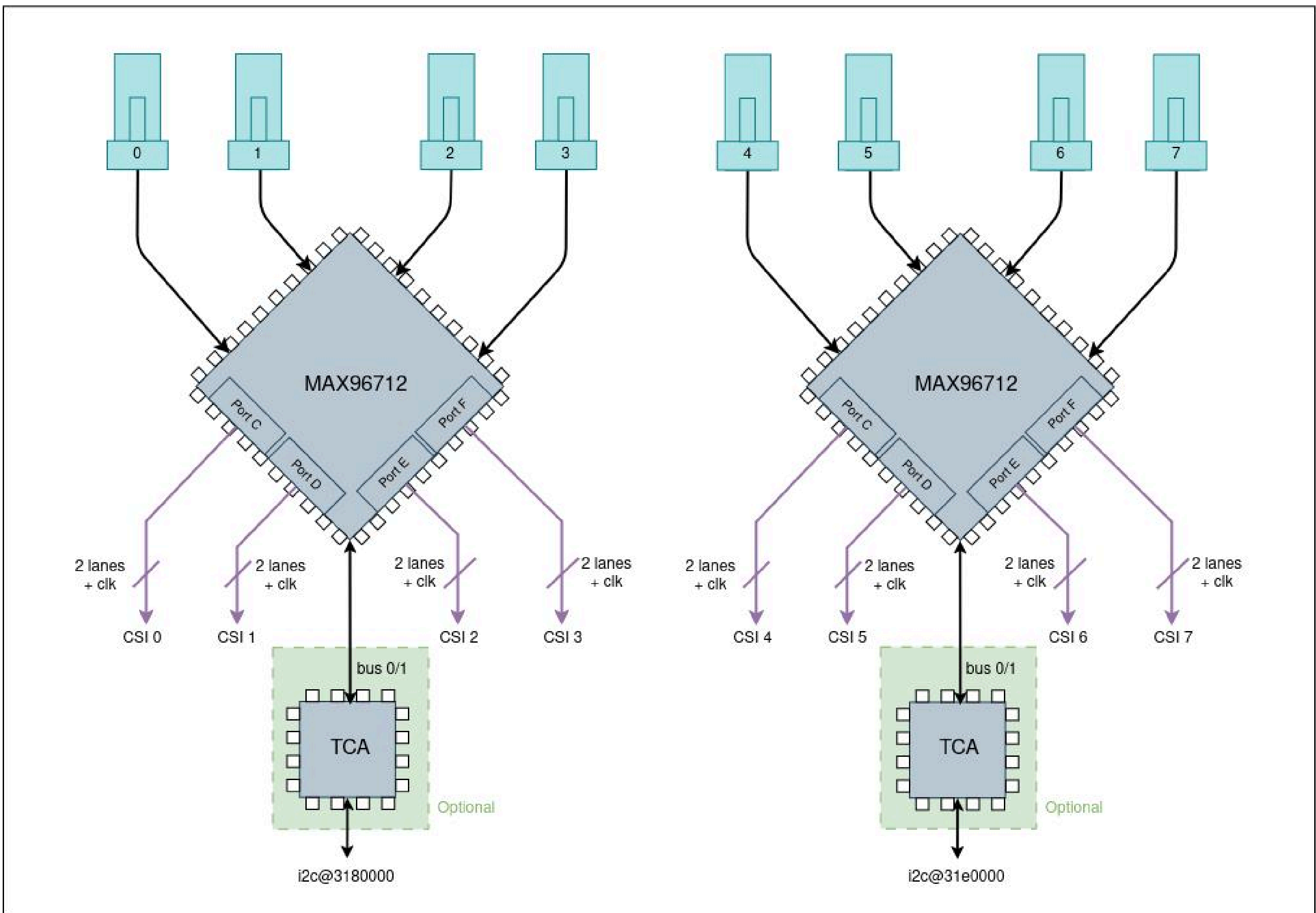


Figure 11: MAX96712 architecture C
Each deserializer has its own I2C port. TCA can help manage i2c address attribution. every MIPI ports are used

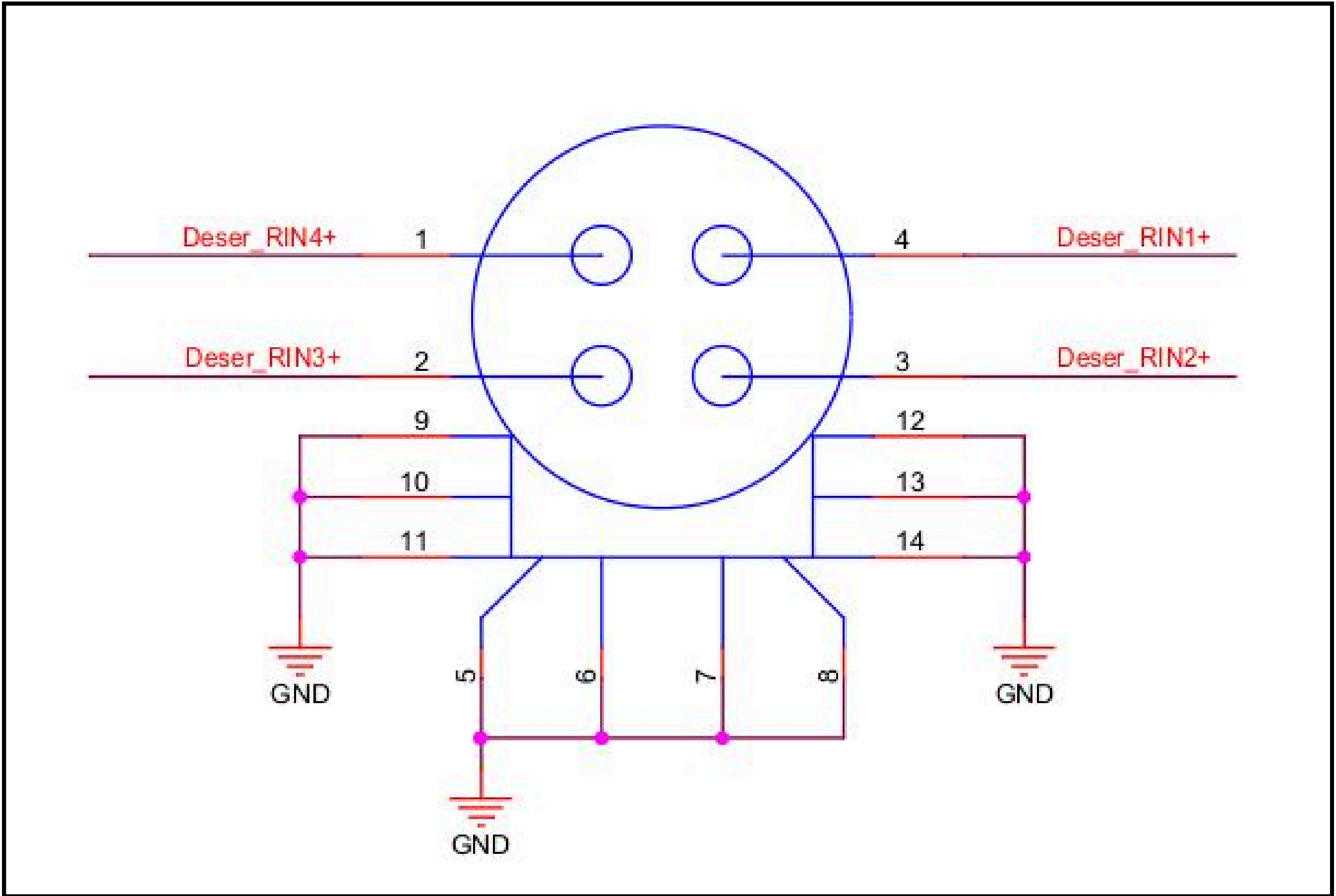
MAX96712 architecture capabilities comparaison			
Architecture	I2c Communication	Bandwidth	Compatibility
A	⚠️ Poor I2C communication	8 ZED X at HD 1200@30fps	AGX, NX, Nano
B	✅	8 ZED X at HD 1200@30fps	AGX, NX, Nano
C	✅	8 ZED X at HD 1200@60fps	AGX

Table 4: MAX96712 architecture capabilities comparaison

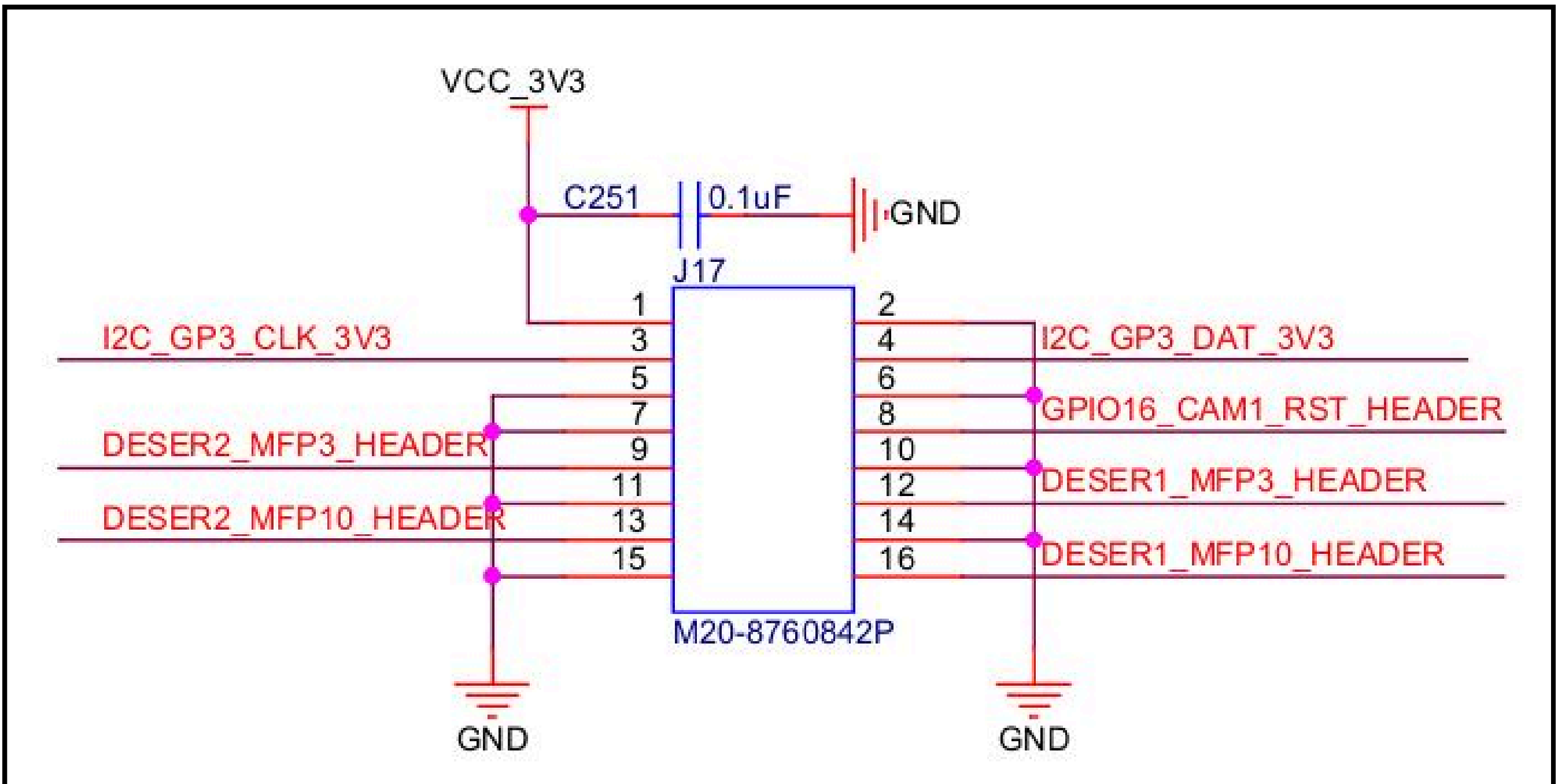
Capture Card Interfaces

For interfacing with the carrier board or other components, examples implemented in the Quad Capture Card are provided below.

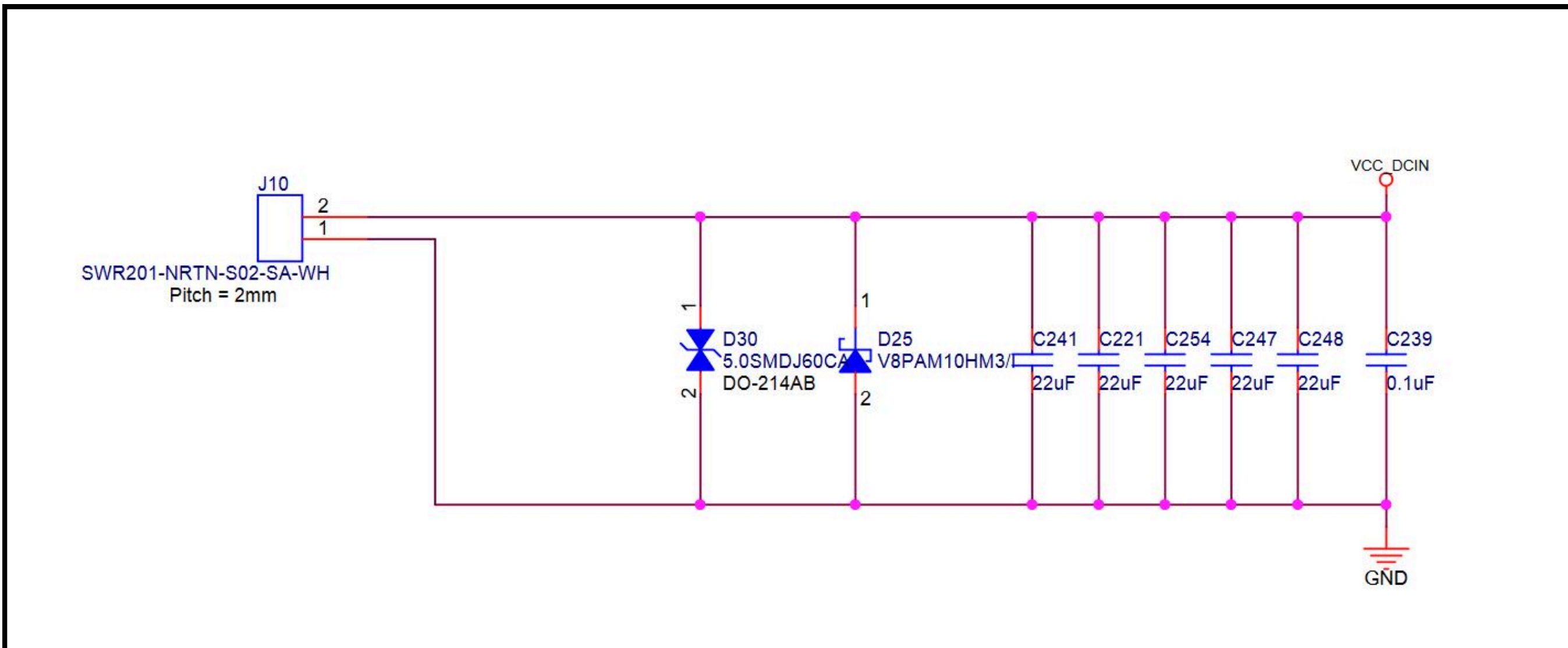
Interface J1 & J2 (Fakra Z 4x1)
2x 4 GMSL2 I/O
Part number TE 1-2304168-9



Interface J17 (GPIO Header)
Part number Harwin M20-8760842P
16 pins 2.54mm pitch



Interface J10 External Power (5V - 19V) - Required
Part number SWR201-NRTN-S02-SA-WH or B2B-PH-K-S
Pitch 2 mm



Capture Card Interfaces

Interface J1 & J2 (Fakra Z 4x1)
2x 4 GMSL2 I/O
Part number TE 1-2304168-9

